

JEE MAIN REVISION MASTER: GASEOUS STATE OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 63 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Gaseous State — Core Principles & Governing Laws	46 PYQs	Core Principle
Very High	Gaseous State — Standard Calculations & Problem Archetypes	15 PYQs	Quantitative Analysis
Moderate	Gaseous State — Applied Cases, Reactions & Traps	3 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Gaseous State — Core Principles & Governing Laws	$PV=nRT$, $\rho = \frac{PM}{RT}$, $c = \sqrt{\frac{\gamma RT}{M}}$
Gaseous State — Standard Calculations & Problem Archetypes	$(p + \frac{a}{V_m^2})(V_m - b) = RT$, $\frac{b}{V_m} = \frac{RT}{p}$. The value of $z = 1 + \frac{p b}{RT} - \frac{a}{p V_m^2 RT}$
Gaseous State — Applied Cases, Reactions & Traps	$z = 1 + \frac{p b}{RT} - \frac{a}{p V_m^2 RT}$, $\rho = \frac{PM}{RT}$. (Nearest Integer) [Given : $R = 0.082$ L atm K ⁻¹ mol ⁻¹ , $C = 100$ mm Hg. The molar mass of geraniol is _____ g mol ⁻¹]

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Gaseous State — Core Principles & Governing Laws (Very High Yield — 46 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Gaseous State.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Gaseous State.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Gaseous State PYQs.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (Numerical): "Three bulbs are filled with CH_4 (4), CO_2 (2) and Ne as shown in the picture. The bulbs are connected through pipes of zero volume. When the..." [Ans: 3]
- [2023 Evening Shift] Q#2 (MCQ): "Arrange the following gases in increasing order of van der Waals constant 'a' A. Ar B. CH_4 (4) C. H_2O D. C_6H_6 Choose the corre..." [Ans: N/A]
- [2023 Morning Shift] Q#2 (Numerical): "The total pressure of a mixture of non-reacting gases X(0.6g) and Y(0.45g) in a vessel is 740mm of Hg . The partial pressure of the ..." [Ans: 555]

3.2 Gaseous State — Standard Calculations & Problem Archetypes (Very High Yield — 15 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Gaseous State.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2025 Morning Shift] Q#1 (MCQ): "At the sea level, the dry air mass percentage composition is given as nitrogen gas: 70.0 , oxygen gas: 27.0 and argon gas: 3.0 . If total pr..." [Ans: N/A]
- [2023 Morning Shift] Q#3 (MCQ): "For 1 mol of gas, the plot of pV vs. p is shown below. p is the pressure and V is the volume of the gas What is the value of compressibility..." [Ans: N/A]
- [2022 Evening Shift] Q#1 (Numerical): "'x' g of molecular oxygen (O_2) is mixed with 200g of neon (Ne). The total pressure of the non-reactive mixture of O_2 and Ne in t..." [Ans: 80]

3.3 Gaseous State — Applied Cases, Reactions & Traps (Moderate Yield — 3 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Gaseous State.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2022 Morning Shift] Q#3 (Numerical): "Geraniol, a volatile organic compound, is a component of rose oil. The density of the vapour is 0.46 gL⁻¹ at 257 °C and 100 mm..." [Ans: 152]
- [2019 Shift] Q#1 (MCQ): "Consider the following table : Gas a/(k Pa dm³ mol⁻¹) b/(dm³ mol⁻¹) A 642.32 0.05196 B 155.21 0.04136 C 431.91 0.05196 D 155.21 0.4382..." [Ans: N/A]
- [2017 Shift] Q#1 (MCQ): "Among the following, the incorrect statement is :..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 7th April Morning Shift (Q#1)

At the sea level, the dry air mass percentage composition is given as nitrogen gas: 70.0, oxygen gas: 27.0 and argon gas: 3.0. If total pressure is 1.15 atm, then calculate the ratio of following respectively: (i) partial pressure of nitrogen gas to partial pressure of oxygen gas (ii) partial pressure of oxygen gas to partial pressure of argon gas (Given: Molar mass of N, O and Ar are 14, 16 and 40 g mol⁻¹ respectively.)

- (A) 5.46, 17.8
- (B) 2.96, 11.2
- (C) 4.26, 19.3
- (D) 2.59, 11.85

Correct Answer: None

Solution Analysis: Partial Pressure Ratio of Nitrogen to Oxygen : The partial pressure of a gas is calculated using its mole fraction. The mole fraction is the ratio of the number of moles of the gas to the total number of moles of all gases. For nitrogen (N₂), the mole fraction $\frac{n_{\text{N}_2}}{n_{\text{N}_2} + n_{\text{O}_2} + n_{\text{Ar}}}$ can be simplified as $(70/28) / (70/28 + 27/32 + 3/40)$. For oxygen (O₂), the mole fraction is $(27/32) / (70/28 + 27/32 + 3/40)$. Therefore, the ratio of the partial pressure of nitrogen to oxygen is : $\frac{P_{\text{N}_2}}{P_{\text{O}_2}}$

Benchmark Problem 2: JEE Main 2023 (Online) 8th April Morning Shift (Q#1)

Three bulbs are filled with CH₄, CO₂ and Ne as shown in the picture. The bulbs are connected through pipes of zero volume. When the stopcocks are opened and the temperature is kept constant throughout, the pressure of the system is found to be _____ atm. (Nearest integer)

Correct Answer: 3

Solution Analysis: Dalton's law of partial pressure states that whenever two or more gases, which do not react chemically, are enclosed in vessel, the total pressure is equal to sum of partial pressure of each gas. From Dalton's partial pressure law, $P_{\text{f}}V_{\text{f}} = P_{\text{1}}V_{\text{1}} + P_{\text{2}}V_{\text{2}} + P_{\text{3}}V_{\text{3}}$ $P_{\text{f}} \times 9 = 2 \times 2 + 4 \times 3 + 3 \times 4$ $P_{\text{f}} = (28/9) = 3.11 = 3$

Benchmark Problem 3: JEE Main 2022 (Online) 29th June Morning Shift (Q#3)

Geraniol, a volatile organic compound, is a component of rose oil. The density of the vapour is 0.46 g L⁻¹ at 257 °C and 100 mm Hg. The molar mass of geraniol is _____ g mol⁻¹. (Nearest Integer) [Given : R = 0.082 L atm K⁻¹ mol⁻¹]

Correct Answer: 152

Solution Analysis: From ideal gas equation we know $PV = nRT \Rightarrow PV = (W/M)RT \Rightarrow P = (W/V) \cdot (RT/M) \Rightarrow P = d \cdot (RT/M)$ [$d = (W/V)$] We know, 760 mm of Hg = 1 atm $\therefore$ 100 mm of Hg = $(100/760)$ atm $\therefore$ Pressure (P) = $(100/760)$ atm Density (d) = 0.46 R = 0.082 L atm K⁻¹ mol⁻¹ T = (257 + 273) K = 530 K Putting the values in above equation, we get $(100/760) = (0.46 \times 0.082 \times 530 / M) \Rightarrow M = 152$